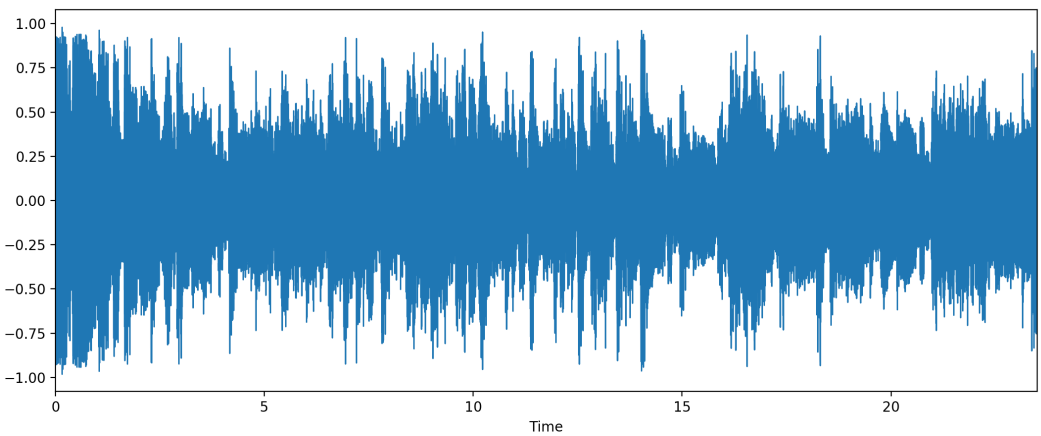
Personal Project: A Brief Inquiry Into Data Science

My inspiration for this project began with my favorite band, The 1975, releasing a new single for their upcoming album. The release of a single from an upcoming album gives fans a glimpse to see if their favorite artist will stay "on brand" or explore a new style of music in the future. I decided to research and take a data science approach to rank the similarity of every previous song from The 1975 against their newest single: "Frail State of Mind". The title of my project pays homage to The 1975's 2018 album: "A Brief Inquiry Into Online Relationships".

I began researching the Librosa library where I was able to produce a plot of a spectrogram from a short clip of a live recording I had. I began to explore different libraries for my analysis because producing the audio waveform was too specific and required me having downloaded access to all the songs in the data set I would be producing.

To the left: Plot of a short audio clip of live music



I then found the Spotipy library from Spotify for Developers, which seemed most ideal for my data collection as I was able to have access to all The 1975's tracks, and the library included methods that output calculations analyzing any track on Spotify.com.

Data Collection

Using the method audio_features from the spotipy library, returns a list of dictionaries containing different measurements per track. For example, the track "Give Yourself a Try" returns:

[{'danceability': 0.334, 'energy': 0.786, 'key': 11, 'loudness': -5.002, 'mode': 1, 'speechiness': 0.0592, 'acousticness': 3.17e-05, 'instrumentalness': 0, 'liveness': 0.517, 'valence': 0.901, 'tempo': 182.933, 'type': 'audio_features', 'id': '4rmIfFUZhhi9sS5IYtpkXw', 'uri': 'spotify:track:4rmIfFUZhhi9sS5IYtpkXw', 'track_href': 'https://api.spotify.com/v1/tracks/4rmIfFUZhhi9sS5IYtpkXw', 'analysis_url': 'https://api.spotify.com/v1/audio-analysis/4rmIfFUZhhi9sS5IYtpkXw', 'duration_ms': 196589, 'time_signature': 4}]

The first 11 values are the ones important for analysis of the tracks and making the comparisons. I want to make a table where the left hand column is the track names, and the row across the top is the keys in the audio features library.

I am familiar with Pandas and NumPy libraries, both of which can be used to produce tables. I will be using pandas for this data.

If you pass `orient='index'`, the keys will be the row labels. In this case, you can also pass the desired column names:

```
In [58]: pd.DataFrame.from_dict(dict([('A', [1, 2, 3]), ('B', [4, 5, 6])]),
   ....:                        orient='index', columns=['one', 'two', 'three'])
   ....:
Out[58]:
   one  two  three
A    1    2      3
B    4    5      6
```

Now that I have a csv of the audio features of all the tracks, I need to use the audio features from the single track I wish to analyze and compare to all the current data to find the one that is most similar.

Data Analysis

The csv file contains all tracks from The 1975's previous albums, and since I wish to analyze the similarity of a single track from a new album, I had to manually add the track "Frail State of Mind" from the unreleased album to the csv file as the last row.

Then I created a function: similarity_calculator, that calculates the percent error between two tracks given a specific feature from audio_features. I use this function to calculate a cumulative error percent for each track across all categories. There is a nested for loop to calculate every track's (rows) category's (columns) percent error.

Since the only categories worth analyzing are ones that contain numerical values, I have it break out of the for loop once it gets to a string. The final percent errors are stored in a new dictionary which can be sorted to find the track with the lowest percent.

Conclusion

The results I produced shows that the track "I Always Wanna Die (Sometimes)" is most similar to "Frail State of Mind", and "An Encounter" is least similar. Based on my initial impression of the song, I would have predicted that it would closely match to one from their recent album: "A Brief Inquiry Into Online Relationships", because I would describe The 1975's sound to have evolved from indie rock to include more electro pop in their newer work. In short, it makes sense to see the first three tracks that were calculated to be most similar to be from their recent album. The least similar tracks included some short tracks that were used as transition pieces in

their self titled album and instrumental tracks. Therefore, the data I produced returned the songs with the most and least similar features with the newest track, as well as displayed the band's evolution of style over time.

There are some limitations to this analysis, one being that each feature was weighted equally as a comparison in analyzing similarity. Analyzing similarity in music could be a little subjective in this area, which is why I decided to weight them equally. For example, if an artist releases an acoustic version of an originally electric song, would you still consider them "similar"? Or two songs could have two very similar beats but one is entirely instrumental, or in a special case of my data set, an extended time period of silence in the track "You". This track's first three minutes or so is a song and the rest is complete silence, a stylistic touch from the artist when they released the track, which consequently skewed all the audio feature values to be close to 0. But overall, I am satisfied with the results I have produced in getting familiar with the spotify library.